

ATEK1801

Programming Supplement

20-550 MHz 32-State USB-Controlled Low Pass Filter

Document revision 1.0 | Python API version 1.0.0

This supplement contains the ATEK1801-specific programming information. It is used together with the **ATEK RF Modules Programming Manual**, which describes the connection, the command set, state reporting and the Python API common to all modules.

Software, example programs and datasheets: <https://github.com/atek-midas/atek-rf-modules>

1 Module Identification

Item	Value
Module name (*IDN? response)	ATEK1801
RF chip (datasheet)	ATEK888P5
Description	20-550 MHz 32-State USB-Controlled Low Pass Filter
States	32 (state 0 ... 31)
Python class	<code>atek.TunableLowPassFilter</code>

RF performance data is found in the datasheet of the RF chip.

2 States

32-band tunable low pass filter. Band n is selected with state n – 1. The front-panel buttons step through the same states.

32 bands. Nominal cut-off frequencies:

Band	State	Cut-off (MHz)	Band	State	Cut-off (MHz)
1	0	27	17	16	150
2	1	28	18	17	155
3	2	29	19	18	160
4	3	30	20	19	164
5	4	31	21	20	168
6	5	32	22	21	172
7	6	33	23	22	176
8	7	34	24	23	180
9	8	48	25	24	230
10	9	51	26	25	245
11	10	55	27	26	265
12	11	58	28	27	285
13	12	68	29	28	310
14	13	75	30	29	350
15	14	95	31	30	425
16	15	112	32	31	550

3 Raw Command Example

Direction	Line	Comment
PC → Module	*IDN?	Identify the module
Module → PC	ATEK1801	
PC → Module	SET:15	Select band 16
Module → PC	STATE:15	Confirmation
PC → Module	GET?	Read the current state
Module → PC	STATE:15	Band 16, cut-off 112 MHz

Command syntax and state reporting are described in Section 3 of the ATEK RF Modules Programming Manual.

4 Python API

4.1 Class atek.TunableLowPassFilter

`atek.connect()` returns an `atek.TunableLowPassFilter` object when a ATEK1801 is connected. In addition to the common methods (ATEK RF Modules Programming Manual, Section 5.5) it provides:

Method / attribute	Description
<code>set_band(band, verify=True)</code>	Selects band 1...32. Returns the band.
<code>get_band()</code>	Returns the active band (1...32).
<code>get_cutoff_mhz()</code>	Returns the nominal cut-off frequency of the active band in MHz.
<code>set_cutoff_mhz(mhz, verify=True)</code>	Selects the lowest band whose cut-off is $\geq$ mhz, i.e. the tightest filter that still passes mhz. Returns the selected cut-off in MHz.
<code>CUTOFFS_MHZ</code>	Tuple of the 32 nominal cut-off frequencies, index = state.

Band numbers start at 1; state indexes start at 0. Frequencies are given in MHz.

4.2 Example program

```
import atek_rf_modules_scpapi as atek

with atek.connect("COM5") as lpf:          # returns an atek.TunableLowPassFilter object
    print("IDN      :", lpf.identify())      # ATEK1801

    lpf.set_band(10)                        # band 10 (state 9)
    print("Band    :", lpf.get_band(), "->", lpf.get_cutoff_mhz(), "MHz")

    fc = lpf.set_cutoff_mhz(100)            # lowest cut-off >= 100 MHz
    print("100 MHz :", "band", lpf.get_band(), "->", fc, "MHz")
```

Output:

```
IDN      : ATEK1801
Band     : 10 -> 51 MHz
100 MHz  : band 16 -> 112 MHz
```